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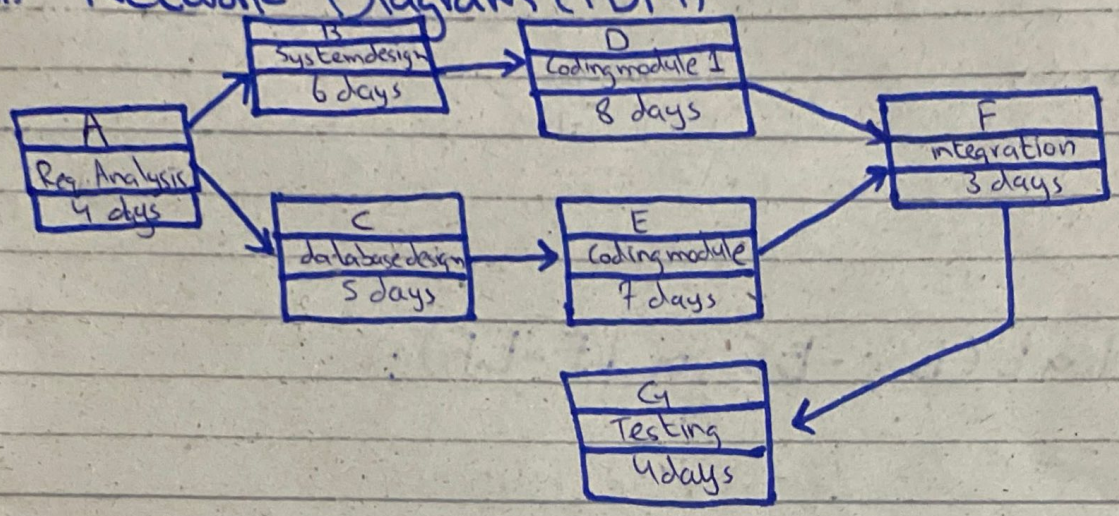
Q1 Activity Network & critical Path:

(i) Network Diagram (PDM)



(ii) ES, EF & LS, LF

Forward Pass (ES, EF):

A: ES = 0 , EF = 0+4 = 4
B: ES = 4 , EF = 4+6 = 10
C: ES = 4 , EF = 4+5 = 9
D: ES = 10 (Only from B), EF = 10+8 = 18
E: ES = 9 (Only from C), EF = 9+7 = 16
F: ES = max{EF of D, EF of E} = max(18, 16) = 18
EF = 18+3 = 21
G: ES = 21 , EF = 21+4 = 25

Project duration = 25 days

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Backward Pass (LS, LF):

G : LF = 25 , LS = 25 - 4 = 21
F : LF = 21 (G's LS) , LS = 21 - 3 = 18
D : LF = 18 (F's LS), LS = 18 - 8 = 10
E : LF = 18 (F's LS), LS = 18 - 7 = 11
B : LF = 10 (D's LS) , LS = 10 - 6 = 4
C : LF = 11 (E's LS) , LS = 11 - 5 = 6
A : LF = min (LS of B , LS of C) = min(4, 6) = 4
LS = 4 - 4 = 0

Float (LS - ES or LF - EF):

A : LS - ES = 0 - 0 = 0
B : 4 - 4 = 0
C : 6 - 4 = 2
D : 10 - 10 = 0
E : 11 - 9 = 2
F : 18 - 18 = 0
G : 21 - 21 = 0

(iii) Critical Path:

Activities with float = 0 : A → B → D → F → G
Critical Path : A - B - D - F - G

(iv) Total Duration:

25 days

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Q2 PERT Analysis:

(i) Expected Duration:

$$te = \frac{O + 4M + P}{6} = \frac{3 + 4(4) + 11}{6} = \frac{30}{6} = 5 \text{ days}$$

Expected Duration = te = 5 days

(ii) Compare with original estimated:

Original = 4 days
PERT Expected = 5 days
Difference = +1 days (25% increase)
PERT analysis shows that when accounting for uncertianity, the expected duration for testing is 1 day longer than the original estimate

(iii) Impact:

(a) Critical Path:

Original Critical Path: A(4) → B(6) → D(8) → F(3) → G(4) = 25 days

PERT expected: A(4) → B(6) → D(8) → F(3) → G(5) = 26 days

Duration increases from 25 - 26 days

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(b) Management Implication

Following areas will be impacted

- Project Buffer
- Critical Path monitoring
- Risk Migration

Q3 Slack/Float Calculation:

(i) Slack/Float = LS - ES

	LS	ES	Slack
A	0	0	0
B	4	4	0
C	6	4	2
D	10	10	0
E	11	9	2
F	18	18	0
G	21	21	0

(ii) Activities with 0 slack:

A, B, D, F, G

These lie on critical Path and cannot be delayed

Activities that can be delayed

Slack > 0 (flexible)

C → 2 days

E → 2 days

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Q4 Cost Estimation

(i) Each activity cost:

Act	Cost	Duration	Total
A	200	4	800
B	300	6	1800
C	250	5	1250
D	400	8	3200
E	350	7	2450
F	500	3	1500
G	300	4	1200

(ii) Total Project Cost:

$12,200

(iii) Identify:

- Direct costs:

Direct cost = cost directly related from A → G ⇒ $12,200

- Indirect costs.

Indirect cost = Not tied to specific activity like QA, training, facilities etc

Q5 Earned value management (EVM).

(i) Cost Variance (CV):

CV = 8000 − 12000 = −4000

CV = − $4000

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- Schedule Variance (SV):

$SV = 8000 - 10000 = -2000$

$SV = -\$2000$

- Cost Performance Index (CPI):

$CPI = \frac{8000}{12000} = 0.67$

- Schedule Performance Index (SPI):

$SPI = \frac{8000}{10000} = 0.8$

(ii) Interpret Results:

- Is Project underbudget or over budget:

CV is -ive

CPI is less than 1

So, Project is over budget

- Ahead or behind Schedule

SV is -ive

SPI is less than 1

So, Project is behind schedule

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